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DS 5030

Lab 1

I tested TF-IDF correlation against the three existing baselines and found that it does not outperform them. Raw dot product scored 0.093, cosine similarity scored 0.191, and TF-IDF cosine scored 0.210. TF-IDF correlation came in at 0.198 which is an improvement over Raw dot product plain cosine, but is still behind TF-IDF cosine, which remains the best-performing method overall. Therefore, my hypothesis that TF-IDF correlation would perform better than TF-IDF cosine was wrong.

The likely explanation here lies in how correlation treats each song's standard deviation. Dividing by a song's own spread amplifies scores for low-variance songs, and a vector with just one or two nonzero, high-IDF words naturally has a small standard deviation. Therefore, as a result, correlation can inflate the similarity score for songs whose pattern is really just one or two rare-word spikes, rather than a genuinely broad song genre or content match.

This is the same weakness correlation that was shown earlier in the lab, back in Step 3, where it was already observed to disproportionately reward low-variance vectors. Layering TF-IDF's rare-word weighting on top of that compounds the issue, as TF-IDF inflates the importance of rare words, and correlation then further amplifies any song whose variance is dominated by a small number of those rare, heavily-weighted terms. The two effects reinforce each other in the wrong direction, which can help explain why this combination underperformed when compared to TF-IDF cosine despite each individual idea (centering and reweighting) working well on its own.